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Research

Effectiveness of Hand Massage with Pandan Lotion on Blood Pressure in Pregnant Women in Soropia District, Konawe Regency

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ABSTRACT

Background: Hypertension in pregnancy can increase the risk of maternal and perinatal complications, making safe and easily applicable supportive interventions relevant to maternal care. Hand massage may promote relaxation, while fragrant pandan (*Pandanus amaryllifolius* Roxb.) can be incorporated into a lotion used as a massage medium.

Methods: This quantitative quasi-experimental study used a two-group pretest-posttest design involving 60 pregnant women with a gestational age of ≥ 20 weeks in Soropia District, Konawe Regency. Thirty participants received hand massage with pandan lotion and 30 received hand massage without pandan lotion. The intervention was administered twice weekly for two months. Descriptive statistics, the Shapiro-Wilk test, paired t-test, and Mann-Whitney test were used for analysis.

Results: In the intervention group, mean systolic blood pressure decreased from 136.20 to 126.53 mmHg and mean diastolic blood pressure from 87.53 to 80.57 mmHg. Within-group changes were significant for systolic blood pressure ($p < 0.001$) and diastolic blood pressure ($p = 0.001$). The control group also showed significant within-group reductions. Between groups, the change in systolic blood pressure was not significantly different ($p = 0.074$), whereas the change in diastolic blood pressure was significantly greater in the intervention group ($p < 0.001$).

Conclusion: Hand massage produced a relaxation-related reduction in blood pressure in both groups, while the addition of pandan lotion was associated with a greater reduction in diastolic blood pressure. This intervention may be considered as supportive complementary care and should not replace clinical monitoring or standard medical therapy for hypertension in pregnancy.

KEYWORDS: Hand massage, pandan lotion, blood pressure, hypertension, pregnant women

INTRODUCTION

Pregnancy and childbirth are natural events that are highly anticipated by mothers and their families. Although most pregnancies proceed safely, complications may still occur and threaten the safety of both the mother and fetus. One important cardiovascular condition during pregnancy is hypertension. Hypertension in pregnant women can increase the risk of complications and is one of the problems contributing to maternal mortality.¹

During pregnancy, blood volume increases to meet the oxygen and nutritional needs of the placenta, uterus, breasts, and maternal tissues. Entering the third trimester, various physical complaints and anxiety about childbirth can increase maternal stress. These psychological conditions may affect blood pressure. One non-pharmacological approach that can help mothers feel more relaxed is prenatal massage. Research in pregnant women with hypertension in pregnancy has shown that prenatal massage can help reduce blood pressure.²

Massage provides stimulation through pressure and rhythmic movements that help improve circulation and create a sense of comfort. In pregnant women, a relaxed state can help reduce tension and support vasodilation. Pregnancy massage has also been reported to produce changes in blood pressure among third-trimester pregnant women.³

The application of massage during pregnancy must continue to consider safe areas and avoid inducing contractions. Massage techniques performed slowly can be used as complementary therapy to reduce discomfort during pregnancy. Such techniques have been applied in the third trimester to help reduce physical complaints.⁴

Indonesia has rich biodiversity that can be utilized as supportive material for non-pharmacological therapy. One commonly found plant is fragrant pandan (*Pandanus amaryllifolius* Roxb.). Pandan leaves contain several compounds, including alkaloids, terpenoids, steroids, flavonoids, saponins, chlorophyll, tannins, glycosides, and essential oils. The aroma of pandan may provide a relaxing sensation, while aromatic materials combined with massage have been investigated as complementary approaches to support changes in blood pressure.⁵

Pandan is also associated with flavonoid content that may have antihypertensive activity. In this study, pandan was used in lotion form as a medium for hand massage. Lotion facilitates massage movements while providing the characteristic aroma of pandan. Clinical observations in hypertensive patients have reported an association between fragrant pandan aroma and changes in blood pressure.⁶

The combination of hand massage and pandan lotion was developed as a simple supportive intervention. Massage is expected to provide relaxation through touch, while pandan lotion serves as a massage medium that may improve comfort. This approach is not intended to replace medical therapy for hypertension in pregnancy, but to complement standard maternal care.

Based on this rationale, this study aimed to describe participant characteristics and blood pressure before and after intervention and to compare changes in systolic and diastolic blood pressure between pregnant women receiving hand massage with pandan lotion and those receiving hand massage without pandan lotion.

MATERIALS AND METHODS

Study design

This quantitative study used a quasi-experimental two-group pretest-posttest design. The intervention group received hand massage using pandan lotion, whereas the control group received hand massage without pandan lotion. Blood pressure was measured before and after the intervention period. Both groups received the intervention twice weekly for two consecutive months.

Population and sample

The population consisted of 152 pregnant women with a gestational age of ≥ 20 weeks in Konawe Regency. Sample size was determined using the Slovin formula with a 10% margin of error, resulting in 60 participants. Sampling used probability sampling with a simple random sampling technique. Thirty participants were included in the intervention group and 30 in the control group.

Inclusion and exclusion criteria

The inclusion criteria were pregnant women with a gestational age of ≥ 20 weeks, blood pressure of 140/90–150/100 mmHg, ability to read and write, and willingness to participate. The exclusion criteria were allergy to pandan lotion and blood pressure $\geq 160/110$ mmHg.

Instruments and procedures

The research instruments included questionnaires on participant characteristics and knowledge of hypertension, a sphygmomanometer for blood pressure measurement, the Hand Massage with Pandan Lotion module as an intervention guide, pandan lotion as the intervention material, and a chair or bed for performing the massage.

Participant characteristics recorded included age, gestational age, gravida, parity, education, and occupation. Blood pressure was measured before and after the two-month intervention period.

Statistical analysis

Univariate analysis was used to summarize participant characteristics and systolic and diastolic blood pressure. Data normality was assessed using the Shapiro-Wilk test. Pretest and posttest blood pressure values within each group were compared using paired t-tests. Because the distributions of individual blood pressure change scores were non-normal, between-group changes were compared using the Mann-Whitney test. Statistical significance was set at $\alpha=0.05$. P-values smaller than 0.001 are reported as $p<0.001$.

RESULTS

Participant characteristics

Participants were 22–33 years old and had gestational ages of 22–36 weeks. Most participants in both groups were 26–29 years old, were in the third trimester, and had completed senior high school. Baseline demographic and obstetric characteristics are summarized in Table 1.

Table 1. Baseline characteristics of pregnant women in the intervention and control groups in Soropia District, Konawe Regency

Characteristic	Category	Intervention, n (%)	Control, n (%)
Age (years)	22–25	10 (33.3)	10 (33.3)
	26–29	12 (40.0)	12 (40.0)
	30–33	8 (26.7)	8 (26.7)
Gestational age	Second trimester (22–27 weeks)	10 (33.3)	10 (33.3)
	Third trimester (28–36 weeks)	20 (66.7)	20 (66.7)
Gravida	1	8 (26.7)	8 (26.7)
	2	11 (36.7)	12 (40.0)
	3	9 (30.0)	8 (26.7)
	4	2 (6.7)	2 (6.7)
Parity	0	8 (26.7)	8 (26.7)
	1	11 (36.7)	12 (40.0)
	2	9 (30.0)	8 (26.7)
	3	2 (6.7)	2 (6.7)
Education	Bachelor's degree	6 (20.0)	6 (20.0)
	Senior high school	20 (66.7)	20 (66.7)
	Junior high school	4 (13.3)	4 (13.3)
Occupation	Teacher	2 (6.7)	2 (6.7)
	Housewife	24 (80.0)	23 (76.7)
	Self-employed	4 (13.3)	5 (16.7)

Source: Primary Data, 2026

Blood pressure outcomes

Descriptive blood pressure statistics are shown in Table 2. Mean systolic and diastolic blood pressure decreased in both groups after the intervention period, with a larger descriptive reduction in the intervention group, particularly for diastolic pressure.

Table 2. Descriptive blood pressure statistics in the intervention and control groups

Group	Statistic	Pretest systolic	Pretest diastolic	Posttest systolic	Posttest diastolic
Intervention	Mean	136.20	87.53	126.53	80.57
	Median	136.00	88.00	126.00	80.00
	SD	4.88	3.09	4.64	2.78

Group	Statistic	Pretest systolic	Pretest diastolic	Posttest systolic	Posttest diastolic
	Range	18	12	18	10
	Minimum	128	82	118	76
	Maximum	146	94	136	86
Control	Mean	136.87	87.87	129.93	85.93
	Median	137.00	88.00	130.00	86.00
	SD	4.95	3.24	9.49	3.97
	Range	18	12	36	14
	Minimum	128	82	110	80
	Maximum	146	94	146	94

Source: Primary Data, 2026

The Shapiro-Wilk test indicated that all pretest and posttest systolic and diastolic blood pressure values in both groups were normally distributed ($p>0.05$). In contrast, all systolic and diastolic change scores were non-normally distributed ($p<0.001$), supporting the use of the Mann-Whitney test for between-group comparisons (Table 3).

Table 3. Shapiro-Wilk normality test for blood pressure measurements and change scores

Measure	Group/time point	n	p-value	Distribution
Systolic BP	Intervention pretest	30	0.516	Normal
	Intervention posttest	30	0.592	Normal
	Control pretest	30	0.405	Normal
	Control posttest	30	0.154	Normal
Diastolic BP	Intervention pretest	30	0.265	Normal
	Intervention posttest	30	0.084	Normal
	Control pretest	30	0.148	Normal
	Control posttest	30	0.176	Normal
Systolic change	Intervention	30	<0.001	Not normal
	Control	30	<0.001	Not normal
Diastolic change	Intervention	30	<0.001	Not normal
	Control	30	<0.001	Not normal

Source: Primary Data, 2026

Within the intervention group, both systolic and diastolic blood pressure decreased significantly after hand massage with pandan lotion (Table 4).

Table 4. Paired t-test results in the intervention group

Blood pressure	Pretest, mean $\pm$ SD (mmHg)	Posttest, mean $\pm$ SD (mmHg)	p-value
Systolic	136.20 $\pm$ 4.88	126.53 $\pm$ 4.64	<0.001
Diastolic	87.53 $\pm$ 3.09	80.57 $\pm$ 2.78	0.001

Source: Primary Data, 2026

The control group also showed significant within-group reductions in systolic and diastolic blood pressure after hand massage without pandan lotion (Table 5).

Table 5. Paired t-test results in the control group

Blood pressure	Pretest, mean $\pm$ SD (mmHg)	Posttest, mean $\pm$ SD (mmHg)	p-value
Systolic	136.87 $\pm$ 4.95	129.93 $\pm$ 9.49	<0.001
Diastolic	87.87 $\pm$ 3.24	85.93 $\pm$ 3.97	<0.001

Source: Primary Data, 2026

Because both groups improved, the additional effect of pandan lotion was assessed by comparing individual change scores. The difference in systolic change was not statistically significant ($p=0.074$), whereas the diastolic reduction was significantly greater in the intervention group ($p<0.001$) (Table 6).

Table 6. Mann-Whitney comparison of blood pressure change between groups

Blood pressure change	Intervention median (min–max), mmHg	Control median (min–max), mmHg	p-value
Systolic	10 (2–12)	1 (0–24)	0.074
Diastolic	8 (1–8)	0 (0–10)	<0.001

Source: Primary Data, 2026

DISCUSSION

The intervention group showed significant within-group reductions in systolic and diastolic blood pressure after hand massage with pandan lotion. This finding is consistent with Afni and Ristica, who reported a reduction in blood pressure among pregnant women with hypertension after prenatal massage.² Similar findings have been reported for pregnancy massage among third-trimester pregnant women.³ A systematic review of non-pharmacological interventions for hypertension in pregnancy also concluded that selected complementary approaches may help reduce blood pressure when used as supportive management.²¹

The reduction in blood pressure after massage may be related to relaxation, reduced tension, and improved comfort. Foot massage has been reported to help reduce blood pressure in patients with hypertension.⁸ Hand massage has also been shown to reduce anxiety and influence vital signs.⁹ A systematic review and meta-analysis of randomized studies in pregnant women reported that foot reflexology massage may reduce anxiety and help stabilize maternal vital signs, supporting a physiological relaxation pathway for massage.²²

The effect of hand massage on cardiovascular responses has also been reported in hypertensive patients receiving combined relaxation therapy.¹⁰ In patients after coronary angiography, classical hand massage affected blood pressure and other vital signs.¹¹ Nevertheless, a meta-analysis found that the magnitude of the blood-pressure response to massage varies across studies.¹²

Blood pressure also decreased significantly in the control group after hand massage without pandan lotion. This finding indicates that massage itself may contribute to the observed physiological response through relaxation and therapeutic touch. Evidence from hand-massage research in anxiety conditions also supports effects on physiological responses.²⁰

The between-group difference in systolic blood pressure change was not statistically significant ($p=0.074$). Therefore, the present data do not demonstrate an additional systolic benefit from pandan lotion beyond hand massage alone. Clinical guidelines emphasize that hypertensive disorders of pregnancy require blood pressure monitoring and management according to severity.¹³ Accurate blood pressure measurement is also essential in this population.¹⁴ Foot massage has been reported to reduce and stabilize blood pressure among pregnant women with gestational hypertension, supporting massage as a complementary intervention rather than a replacement for standard care.²³

In contrast, the between-group difference in diastolic change was significant ($p<0.001$). The median diastolic reduction was 8 mmHg in the intervention group and 0 mmHg in the control group, suggesting that the combination of hand massage with pandan lotion produced a more pronounced diastolic response. Studies involving fragrant pandan in hypertensive patients have also reported reductions or downward trends in blood pressure.^{7,15} Similar changes have been reported in pregnant women with hypertension receiving foot massage combined with warm-water immersion.²⁴

The use of pandan in this study was based on its local availability, characteristic aroma, and reported secondary-metabolite content. Fragrant pandan extract has been investigated for safety characteristics,¹⁷ and pandan extract has also been formulated in lotion preparations.¹⁸ These data provide a practical basis for its use as a massage medium, although they do not establish a specific pharmacological contribution of the lotion in the present study.

Other non-pharmacological therapies for pregnant women with preeclampsia, including warm-water and lemongrass foot soaking, have been reported to produce changes in blood pressure.¹⁶ Evidence from pregnant women with preeclampsia also showed reductions in systolic and diastolic blood pressure during a foot-massage intervention.²⁵ Such approaches should be considered adjunctive and must be adapted to the clinical condition of the pregnant woman.

Additional related studies provide further context for the present findings. Swedish massage in patients with hypertension and foot massage in postoperative patients have both been reported to influence blood pressure, supporting the plausibility of massage-related cardiovascular responses.^{26,27} In pregnancy-induced hypertension, Benson relaxation therapy and structured relaxation techniques have also been associated with reductions in blood pressure and stress.^{28,29}

Other non-pharmacological interventions in pregnant women with hypertension have shown similar supportive effects. Music therapy and guided imagery have been reported to reduce blood pressure and/or stress among women with pregnancy-induced hypertension, while aerobic exercise during pregnancy has been described as beneficial for

blood-pressure control.³⁰⁻³² A cross-sectional study in pregnant women further highlights the clinical relevance of hypertension and its maternal-fetal complications.³³

Regarding the pandan component, studies published in RJPT have examined *P. amaryllifolius* in herbal preparations and characterized its secondary metabolites.^{34,35} These publications support the biological and formulation rationale for investigating pandan-based products; however, they do not establish that pandan lotion independently lowers blood pressure, and the present between-group findings should therefore be interpreted cautiously.

This study has several limitations. It used a quasi-experimental design, included a relatively small sample from one area, and lacked detailed standardization of the pandan lotion and massage procedure. Information on antihypertensive medication use, detailed blood-pressure measurement procedures, and repeated measurements within each visit was also unavailable. These factors may influence the observed effects. Blood-pressure targets during pregnancy should continue to be guided by appropriate clinical monitoring.¹⁹ Future studies should use larger samples, standardized lotion composition and massage protocols, repeated blood-pressure measurements, and explicit control of concomitant antihypertensive therapy.

CONCLUSION

Hand massage with pandan lotion was associated with significant within-group reductions in systolic and diastolic blood pressure. However, when changes were compared between groups, the additional effect was not statistically significant for systolic blood pressure ($p=0.074$) and was significant for diastolic blood pressure ($p<0.001$). These findings support hand massage with pandan lotion as a potential complementary intervention, particularly for relaxation and diastolic blood pressure control, while clinical monitoring and standard medical therapy for hypertension in pregnancy remain essential.

CONFLICT OF INTEREST

The authors declare no conflict of interest related to this study.

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ETHICAL APPROVAL

This study was conducted after obtaining ethical approval from Kendari Ministry of Health Polytechnic, Kendari, Indonesia. The study protocol was approved under Ethical Clearance No. 12.199/KOMETIK/POLTEKESKDI/XII/2025.

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